Weather Data Aggregator

(WeatherScope)

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1. Problem Statement

The Problem:

- Weather data is scattered across platforms.
- Hard to compare temperatures between multiple cities.
- Users cannot easily calculate statistics **(average, highest, lowest)**.

Requirement:

- Fetch **real-time weather data** for 5 cities.
- Compute **average, highest, and lowest temperatures**.
- Present results **visually and clearly**.

2. The Solution:

<u>Weather Data Aggregator(WeatherScope) Dashboard:</u>

- Fetches data via **Open-Meteo API**.
- Uses **parallel API calls** for efficiency (`Promise.all`).
- Displays **city-wise temperature cards**.
- Generates **bar chart comparison**.
- Shows **map markers** for each city.

Example Output Table:

City	Temperature
Mumbai	*30°C*
Delhi	*28°C*
Kolkata	*31°C*
Bangalore	*26°C*
Chennai	*29°C*

3. Working

Workflow:

User clicks "Fetch Weather Data"

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JavaScript sends API requests

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Parallel fetching using Promise.all()

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API returns JSON

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Extract temperatures & store in array

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Compute stats: Average, Highest, Lowest

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Display results: Cards, Chart, Map

Aggregation Formulas:

Calculation	Formula
Average Temperature	*sum(temps) / number of cities*
Highest Temperature	*math.max(...temps)*
Lowest Temperature	*math.min(...temps)*

4.Conclusion

Outcome:

- Retrieves **real-time weather data** efficiently.
- Implements **asynchronous JS & parallel API calls**.
- Performs **statistical analysis** automatically.
- Displays results via **interactive charts and maps**.

Learning / Future Improvements:

- Add **more cities or live news integration**.
- Include **forecast predictions**.
- Improve **visual presentation and responsiveness**.

Thank You.